

Problem 1

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Problem 1 The following argument shows

$$\lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} = \frac{135 - 4\sqrt{3}}{\sqrt{156}}.$$

State which limit law is used to justify each step. (A particular step may have more than one limit law as a justification.)

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} &= \frac{\lim_{x \rightarrow 3} (5x^3 - 4\sqrt{x})}{\lim_{x \rightarrow 3} \sqrt{x^5 - 87}} \\ &= \frac{\left(\lim_{x \rightarrow 3} 5 \right) \left(\lim_{x \rightarrow 3} (x^3) \right) - \left(\lim_{x \rightarrow 3} 4 \right) \left(\lim_{x \rightarrow 3} \sqrt{x} \right)}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} \\ &= \frac{5 \lim_{x \rightarrow 3} (x^3) - 4 \lim_{x \rightarrow 3} \sqrt{x}}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} \\ &= \frac{5(\lim_{x \rightarrow 3} x)^3 - 4\sqrt{3}}{\sqrt{\lim_{x \rightarrow 3} (x^5) - \lim_{x \rightarrow 3} (87)}} \\ &= \frac{5(3)^3 - 4\sqrt{3}}{\sqrt{3^5 - 87}} \\ &= \frac{135 - 4\sqrt{3}}{\sqrt{156}} \end{aligned}$$

Explanation.

$$\begin{aligned}
 \lim_{x \rightarrow 3} \frac{5x^3 - 4\sqrt{x}}{\sqrt{x^5 - 87}} &= \frac{\lim_{x \rightarrow 3} (5x^3 - 4\sqrt{x})}{\lim_{x \rightarrow 3} \sqrt{x^5 - 87}} && \text{Quotient law} \\
 &= \frac{\left(\lim_{x \rightarrow 3} 5 \right) \left(\lim_{x \rightarrow 3} (x^3) \right) - \left(\lim_{x \rightarrow 3} 4 \right) \left(\lim_{x \rightarrow 3} \sqrt{x} \right)}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} && \text{Product law, Composition} \\
 &= \frac{5 \lim_{x \rightarrow 3} (x^3) - 4 \lim_{x \rightarrow 3} \sqrt{x}}{\sqrt{\lim_{x \rightarrow 3} (x^5 - 87)}} && \text{Continuity} \\
 &= \frac{5(\lim_{x \rightarrow 3} x)^3 - 4\sqrt{3}}{\sqrt{\lim_{x \rightarrow 3} (x^5) - \lim_{x \rightarrow 3} (87)}} && \text{Difference, Continuity} \\
 &= \frac{5(3)^3 - 4\sqrt{3}}{\sqrt{3^5 - 87}} && \text{Continuity} \\
 &= \frac{135 - 4\sqrt{3}}{\sqrt{156}}
 \end{aligned}$$